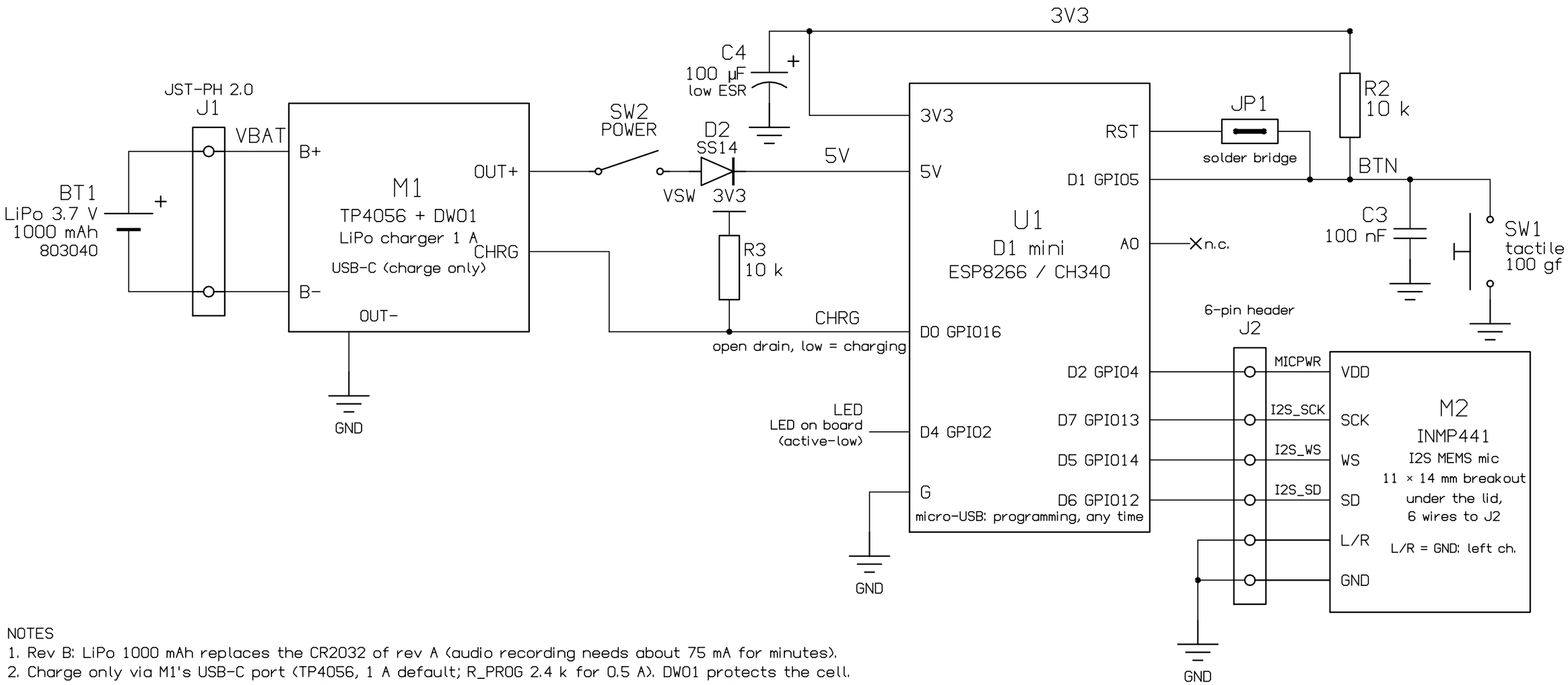



NOTES

1. Rev B: LiPo 1000 mAh replaces the CR2032 of rev A (audio recording needs about 75 mA for minutes).
2. Charge only via M1's USB-C port (TP4056, 1 A default; R_PROG 2.4 k for 0.5 A). DW01 protects the cell.
3. D2 blocks USB back-feed into the LiPo: USB programming is allowed at any time, with SW2 in any position.
4. A0 n.c.; I2S mic on GPIO12/13/14 (the ESP8266's fixed I2S RX pins): SCK = D7, WS = D5, SD = D6, L/R to GND (left channel).
5. M2 VDD power-gated by D2/GPIO4 (1.4 mA). M1 CHRG (open drain, low = charging) -> D0/GPIO16, R3 10 k pull-up to 3V3.
6. JP1 (solder bridge) closed = battery mode: a press pulls RST low and wakes U1 from deep sleep. Open = always_on (USB).
7. GPIO2 drives the LED already on the D1 mini (active-low): on while recording. SW1 pressed = GPIO5 LOW.
8. Rev B.1: M2 INMP441 (11 x 14 mm) sits in a pocket under the lid, sound port over a 2 mm hole; 6 wires to header J2.

| ESP8266 D1 mini MQTT battery button | | |
|---|---|---|
| Rev B.3: LiPo, button, I2S mic<br>Date: 2026-10-26 | Rev B.3<br>CADCraft | Sheet<br>1 / 1 |